TITANIC SURVIVAL PREDICTION USING MACHINE LEARNING

Name: K Sandeep

College Name: Madanapalle Institute Of Technology & Science

Course / Branch:B-Tech/CSE-Data Science

Bootcamp Name: DevTown Bootcamp

Date of Submission:20/04/26

1. Introduction

This project focuses on predicting whether a passenger survived the Titanic disaster using machine learning techniques. It involves data preprocessing, visualization, and model building.

2. Objective

- Analyze the Titanic dataset
- Perform data preprocessing
- Build a predictive model
- Evaluate model performance

3. Dataset Description

The dataset contains features such as Age, Sex, Pclass, Fare, SibSp, Parch, and Embarked. The target variable is Survived (0 = No, 1 = Yes).

4. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

5. Methodology

Data Cleaning:
- Filled missing values
- Removed unnecessary columns

Feature Engineering:
- Created FamilySize

Visualization:
- Plotted graphs

Model Building:
- Logistic Regression

Evaluation:
- Accuracy and Confusion Matrix

6. Results

Accuracy Achieved: 81.01_%

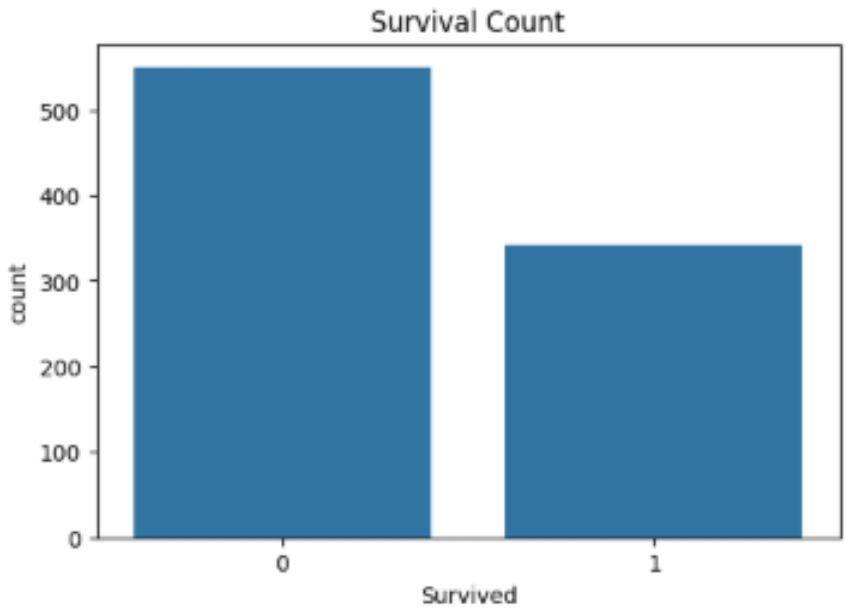
Observations:
- Female passengers had higher survival rate
- Passenger class influenced survival

7. Conclusion

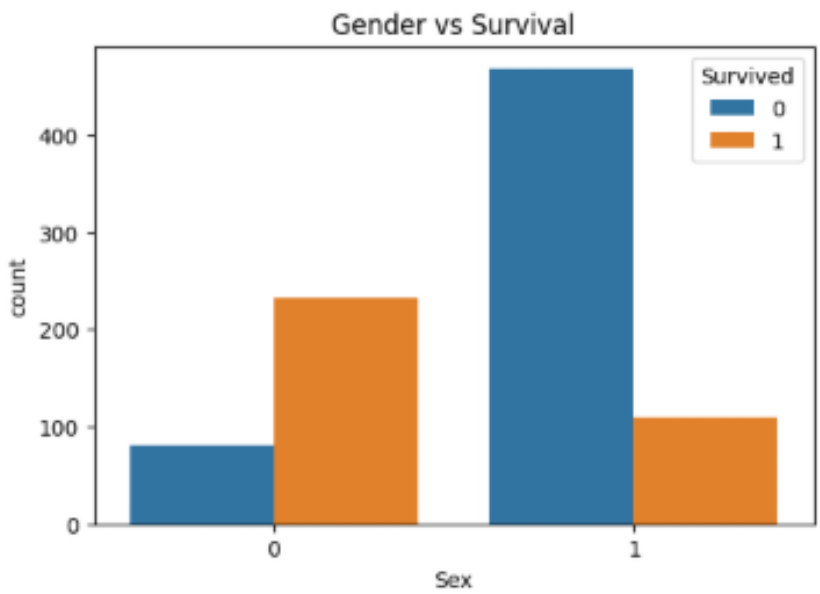
This project demonstrates the complete machine learning workflow including preprocessing, visualization, feature engineering, model building, and evaluation.

8. Screenshots

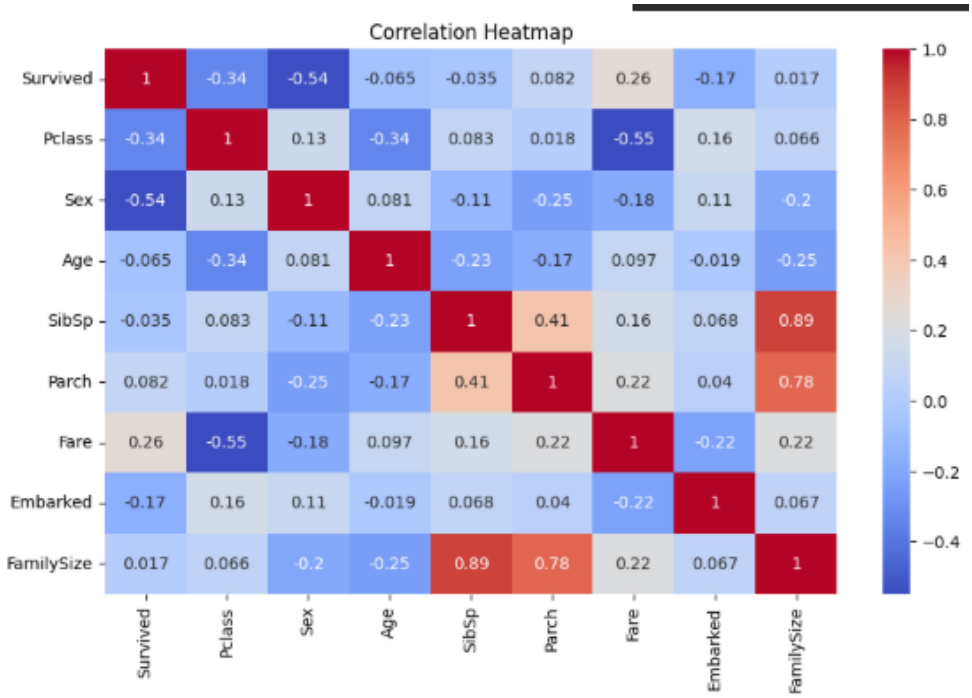
- Survival Count Plot



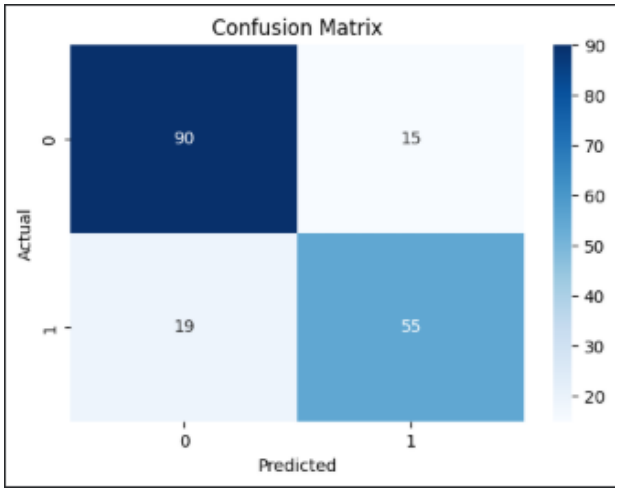
- Gender vs Survival Plot



- Heatmap



- Confusion Matrix



9. References

Titanic Dataset (Kaggle)
Scikit-learn Documentation

10. Declaration

I hereby declare that this project is my own work and has not been copied from any source.